Envision Energy

Design Calculation Description for Sri Lanka 10M/40MWh project

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V1.0

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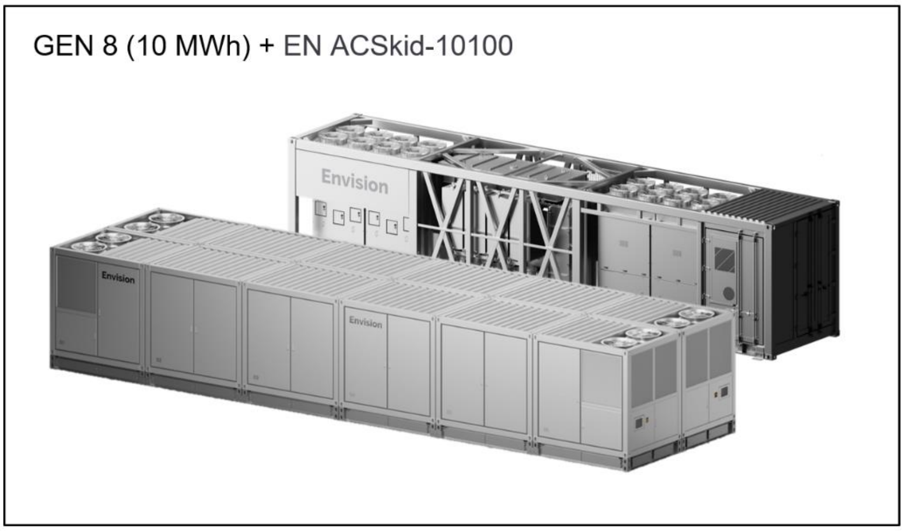
1 Introduction

This document describes design calculations for the Sri Lanka 10M/40MWh BESS project.

2 Key Project Information

Project name	Sri Lanka 10M/40MWh BESS project
Environmental conditions	-30˚C to +45˚C.
Auxiliary losses	Calculated at 35°C
Voltage at Point of Connection (POC)	33kV
Power output at Point of Connection (POC)	10 MW (nominal)
Usable energy at Beginning of Life (BoL)	40 MWh required / 40.2 MWh offered
Cycles per day	1.1
Proposed solution	1 sets ENS-D10G-20100-10100-000 , 1 sets ENS-D06G-24120-10100-000

3 General BESS Design Parameters



AC skid	PCS rated power	4 x 2520 kW
	Transformer voltages	0.69 kV / 33 kV
	Transformer type	3 windings
	Transformer rated capacity	10100 kVA
DC container	Voltage range of DC system	1165 VDC to 1500 VDC
	Rack configuration	10 racks per DC container

4 Proposed BESS Design

4.1 Proposed Overall Solution

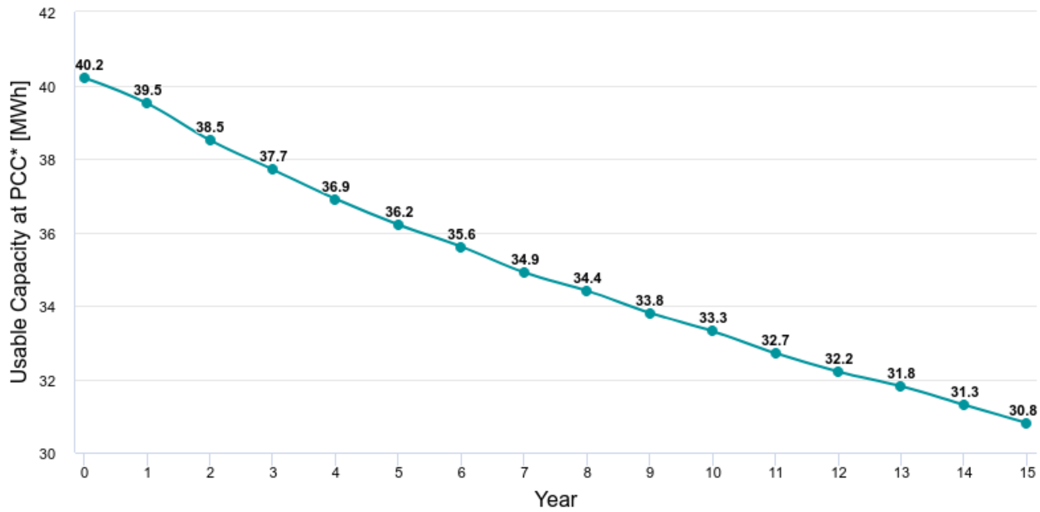
1 set ENS-D10G-20100-10100-000 , 1 set ENS-D06G-24120-10100-000

Item	Value	Unit	Comments
Rated power	15.1	MVA	
Required active power	10	MW	At PoC
Usable active power	14	MW	At PoC
Installed capacity	44.2	MWh	
Usable capacity @ POC	40.2	MWh	At BoL

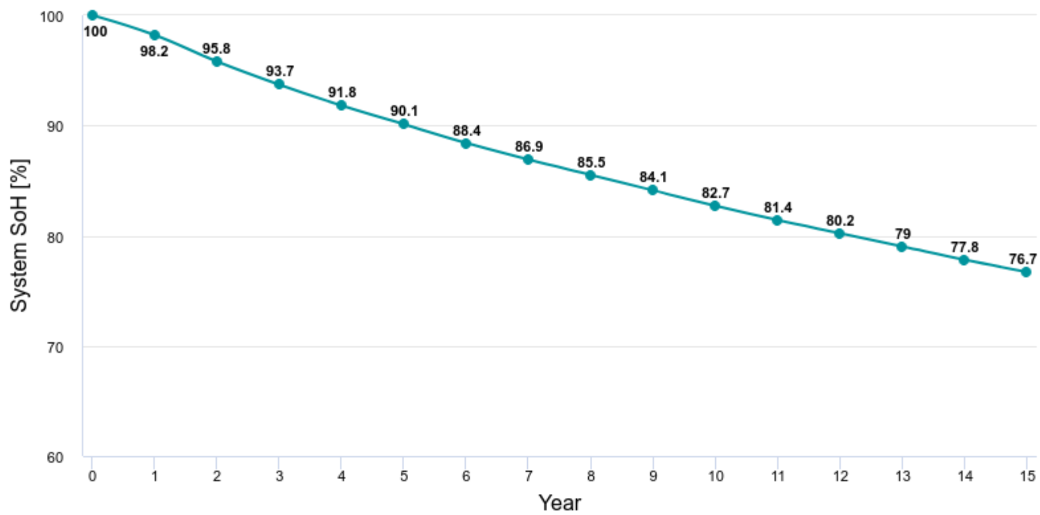
4.2 Performance Curves

1.1 cycles/day										
Year	Usable capacity at 33 kV including aux power (MWh)	Usable capacity at 33 kV excluding aux power (MWh)	System SoH including aux power	System SoH excluding aux power	RTE including aux power	RTE excluding aux power	DC Usable Discharge Capacity (MWh)	DC-DC RTE	Export in Previous Year (MWh)	Cumulative Export (MWh)
BoL	40.2	40.8	100%	100%	86.8%	89.6%	41.9	94.8%		
1	39.5	40.1	98.2%	98.3%	86.6%	89.5%	41.2	94.7%	16008	16008
2	38.5	39.2	95.8%	95.9%	86.4%	89.4%	40.2	94.6%	15675	31684
3	37.7	38.3	93.7%	93.8%	86.2%	89.3%	39.4	94.5%	15310	46994
4	36.9	37.6	91.8%	92%	86.1%	89.2%	38.6	94.4%	14991	61986
5	36.2	36.9	90.1%	90.3%	86%	89.1%	37.9	94.3%	14700	76686
6	35.6	36.2	88.4%	88.7%	85.9%	89.1%	37.2	94.2%	14429	91115
7	34.9	35.6	86.9%	87.2%	85.8%	89%	36.6	94.1%	14174	105290
8	34.4	35	85.5%	85.7%	85.6%	88.9%	36	94.1%	13933	119224
9	33.8	34.4	84.1%	84.3%	85.5%	88.9%	35.4	94%	13702	132926
10	33.3	33.9	82.7%	83%	85.4%	88.8%	34.8	93.9%	13481	146408
11	32.7	33.4	81.4%	81.7%	85.4%	88.7%	34.3	93.9%	13269	159678

12	32.2	32.9	80.2%	80.5%	85.3%	88.7%	33.8	93.8%	13065	172743
13	31.8	32.4	79%	79.3%	85.2%	88.6%	33.3	93.8%	12867	185610
14	31.3	31.9	77.8%	78.2%	85.1%	88.6%	32.8	93.7%	12676	198287
15	30.8	31.5	76.7%	77%	85%	88.5%	32.3	93.6%	12490	210777



*Usable capacity at PCC including the auxiliary losses



*Note that the above values are contingent upon the assumed loss factors shown in the following table. Furthermore, it is important to note that these values may exhibit a slight variation of 1-2 % in response to actual site conditions, attributable to fluctuations in ambient temperature (±5 °C deviation) and the measurement accuracy of external metering equipment (such as PT, CT, etc.). Rest assured, we remain committed to ensuring precision and transparency throughout this project's execution. Furthermore, if the RTE is tested using an electricity meter, we consider that the meter has an error of around 0.5%; therefore, the RTE is deemed compliant within this margin of error.

4.3 Design Sizing

4.3.1 DC Unit

ENS-D10G-20100-10100-000	Value	Unit	Comments
DC container configuration	10051	kWh	10 racks connected in parallel per Container
Nameplate capacity type 1	**20.101**	**MWh**	2 containers connected per AC twin-skid

ENS-D06G-24120-10100-000	Value	Unit	Comments
DC container configuration	6030	kWh	6 racks connected in parallel per Container
Nameplate capacity type 2	**24.121**	**MWh**	4 containers connected per AC twin-skid

4.3.2 AC Skid

Item	Value	Unit	Comments
Step up transformer	10.1	MVA	
PCS	2.52	MVA	
AC unit	10.08	MW	MV AC container. 4x PCS & 1x transformer

4.4 Assumptions

Item	Efficiency	Remarks
Calendar degradation	97 %	From cell manufacture to Site Acceptance Test (SAT). Assuming 6 months from Factory Acceptance Test (FAT) to SAT
Usable DC capacity ratio	98 %	Includes battery cell efficiency & DOD limitations
LV DC cables	99.9 %	Assuming that the cables are aluminium @ 30m, provided by BoP
PCS	98.5%	At Rated Power, provided by Envision
LV/MV transformer	99.2%	Compliance with EU Ecodesign standards, provided by Envision
MV/HV transformer	100 %	Assumed, provided by BoP
MV cable	99.6 %	Assumed, provided by BoP
HV cable	100 %	Assumed, provided by BoP

Item	Energy consumption	Remarks
Aux consumption at BoL	0.16 MW x 4.02 h	Auxiliary consumption cannot be guaranteed separately

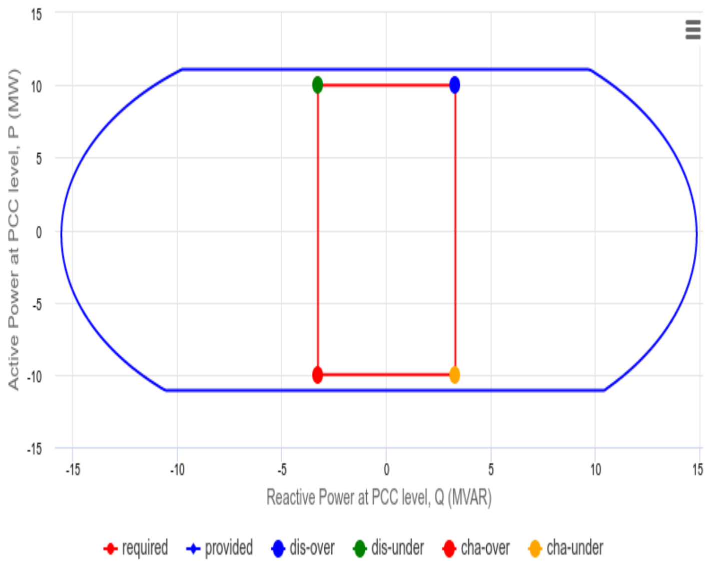
Item	Impedance	Remarks
LV/MV transformer	9 %	Compliance with EU Ecodesign standards
MV/HV transformer	16 %	Assumed, provided by BoP

5 PCC Requirements

Parameter	Unit	Discharge-Overexcitation	Discharge-Underexcitation	Charge-Overexcitation	Charge-Underexcitation
P required at PCC	MV	10	10	-10	-10
S required at PCC	MVA	10.53	10.53	10.53	10.53
Q required at PCC	MVAR	3.29	-3.29	3.29	-3.29
Power factor at PCC	-	0.95	-0.95	0.95	-0.95
P required at PCS	MW	10.33	10.33	-9.67	-9.67
S required at PCS	MVA	11.23	10.56	10.61	9.92

Q required at PCS	MVAR	4.41	-2.17	4.37	-2.21
Power factor at PCS	-	0.92	-0.98	0.91	-0.97

PQ Required and Provided By PCS at PCC Level at 0.9 PU



*Note that the above power flow calculation results are only valid when the PCS terminal voltage is within 0.9–1.1 pu range.

6 Conclusion

This document describes and proposes 1 set ENS-D10G-20100-10100-000, 1 set ENS-D06G-24120-10100-000 for the Sri Lanka 10M/40MWh BESS project. It outlines the essential parameters of the DC unit of the BESS with representative images of the DC containers, including details of the performance curves, State of Health (SoH) of battery and Round-Trip Efficiency (RTE) of the battery in accordance with the project's stipulated requirement. The yearly usable energy capacity of the system has also been presented in the performance curve table throughout the design life of the project.

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